

Futon-the GUI administrator panel of CouchDB

Futon is the Web based GUI panel that is built for CouchDB. It provides the basic interface for a majority of the functions including creation, deletion, updating and viewing of documents. It provides access to the configuration parameters and an interface for initiating replication.

Interaction of CouchDB with Python

Python is a popular programming language that is both general-purpose and cross-platform. It can be used as a standalone suite or a Web based implementation with the APIs.

To interface Python with CouchDB, a specialised package, *couchdb*, is used with the following main modules:

- *couchdb.client* – A client library for interfacing with CouchDB
- *couchdb.mapping* – This module provides the advanced mapping between the JSON documents of CouchDB and Python objects
- *couchdb.view* – This module implements the view based server for the views written in Python

```
>>> import couchdb
```

```
Import CouchDB
```

```
>>> couch = couchdb.Server()
```

This function *couchdb.Server()* creates the *Server Object*.

On live servers, it will be like:

```
>>> couch = couchdb.Server('http://www.myportal.com:5984/')>>> db.delete(doc)
```

```
>>> db = couch.create('test')
```

To create a new database, type:

```
>>> db = couch['mydb']
```

To use the existing database, type:

```
>>> doc = {'Country': 'India'}
```

After selection of a database, create a document and insert it into the database:

```
>>> db.save(doc)
```

The *save()* method returns ID and “rev” for the currently created document.

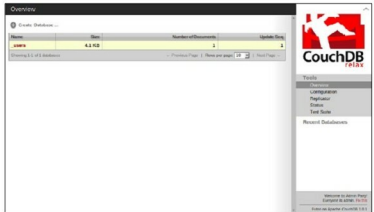


Figure 1: Futon

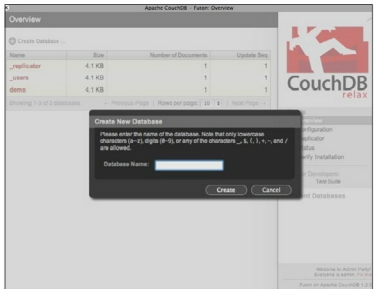


Figure 2: Creating a database in the Futon panel

```
>>> db.delete(doc)
>>> couch.delete('test')
```

The above cleans the document and database.

Couchdbkit: The framework for Python

The goal of *couchdbkit* is to provide a dedicated framework for the Python application to manage and access CouchDB.

The following features are inherent in *couchdbkit*:

- The HTTP backend is used through *py-restclient*
 - Documents are managed dynamically
 - Threadsafeview
 - Design docs are attached with the application and sent to CouchDB
 - Manages documents with a dynamic schema
- You can install *couchdbkit* using Pip as follows:

```
$ curl -O http://python-distribute.org/distribute_setup.py
```